

Exotics

(1. American Option and Stopping Times)

European Pricing (Recap)

Consider a binomial tree with n periods. Let $V_n = g(S_n)$ be the payoff of a derivative security. For example, for call option the payoff $g(S_n) = (S_n - K)^+$

By backward induction, the price of the European Option at any node is given as

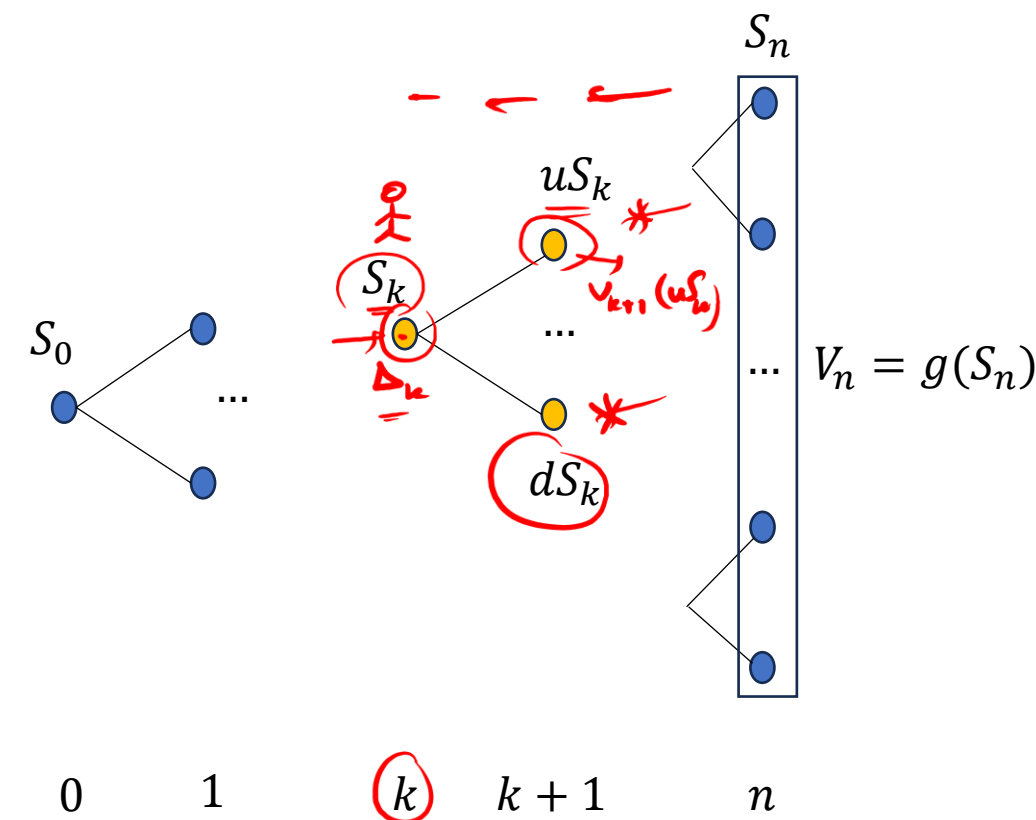
$$v_n(S_n) = g(S_n)$$

$$v_k(S_k) = e^{-r\delta t} [q v_{k+1}(uS_k) + (1-q) v_{k+1}(dS_k)]$$

The hedging portfolio is given as

$$\Delta_k = \frac{v_{k+1}(uS_k) - v_{k+1}(dS_k)}{S_k(u - d)}$$

(Binomial Tree for Stock)



American Pricing

An American option can be exercised early. This implies the derivative price at any time k is at least the exercise value at that point.

$\textcircled{k}$
 $\textcircled{S_k}$

$$\boxed{v_k(S_k)} \geq g(S_k)$$

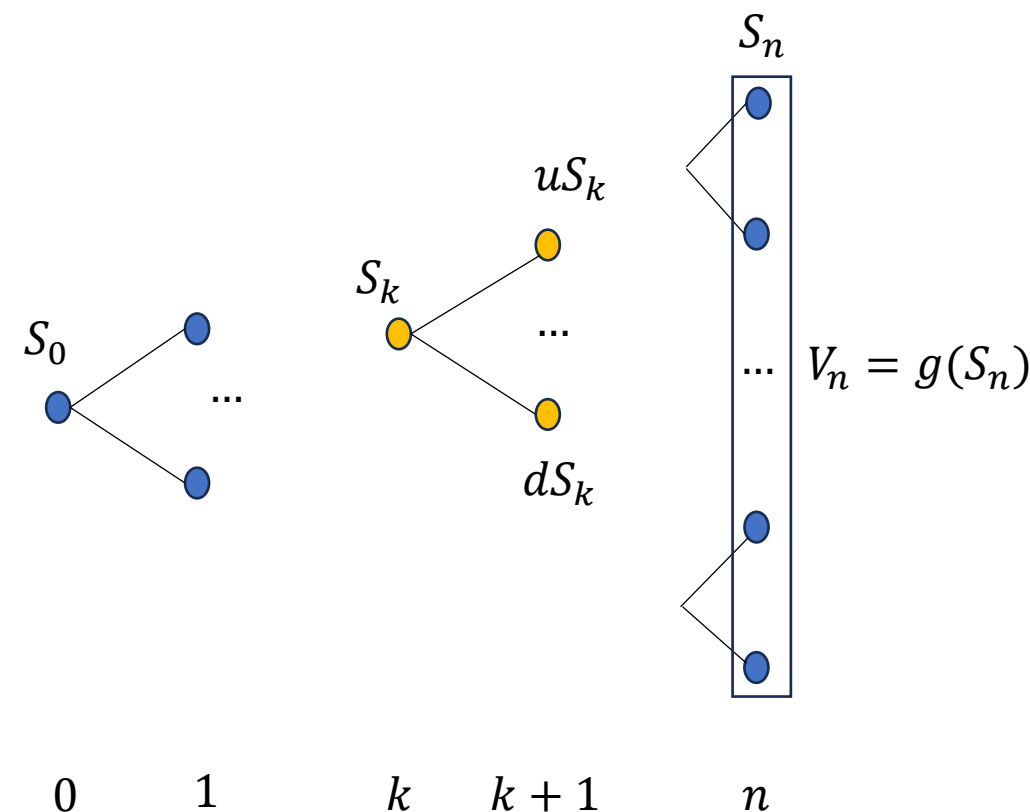
$g(S_k) = (S_k - K)$
 intrinsic value

So, we re-define the backward induction equations as detailed below

$$\boxed{v_k(S_k)} = \max \left\{ \underbrace{e^{-r\delta t} [q v_{k+1}(uS_k) + (1-q) v_{k+1}(dS_k)]}_{\text{(continuation value)}}, \underbrace{g(S_k)}_{\text{(Intrinsic value)}} \right\}$$

$v_n(S_n) = g(S_n) \rightarrow$

(Binomial Tree for Stock)



Exercise

Assume the following :

$S_0 = 4$, $u = 2$, $d = \frac{1}{2}$, One period discounting = $\frac{4}{5}$, Risk neutral probabilities : $q = \frac{1}{2}$, Strike : $K = 5$

- ▶ Find the value of a European Put and an American put using a two-period binomial tree ($n = 2$).
- ▶ Find out the hedging portfolio using the formula developed previously.
- ▶ Verify at every node if the replicating portfolio matches the derivative.

$$\begin{array}{cc}
 \Delta, 4 & (\cancel{\Delta}, 4) \\
 \downarrow \quad \downarrow & \downarrow \quad \downarrow \\
 S & B \\
 S & B
 \end{array}$$

$$\boxed{V} = \Delta \cdot S + 4 \cdot B$$

Solution



Put Option

$S_0 =$	4
$K =$	5
$DF =$	0.8
$q =$	0.5
$u =$	2
$d =$	0.5

Stock Price Tree

		16
	8	
4		4
	2	
		1

Bond Price Tree

		1.5625
	1.25	
1		1.5625
	1.25	
		1.5625

$v_n(S_n) = g(S_n)$

$v_k(S_k) = e^{-r\delta t}[qv_{k+1}(uS_k) + (1 - q)v_{k+1}(dS_k)]$

European Put

		0
	0.4	
0.96		1
	2	
		4

Hedging Portfolio : Δ

	$\Delta_1(up)$	
	-0.0833	
Δ_0		
-0.2667		
	-1	
	$\Delta_1(down)$	

Bank account : ψ

	0.85333	
2.02667		
	3.2	

$v_n(S_n) = g(S_n)$

$v_k(S_k) = \max\{e^{-r\delta t}[qv_{k+1}(uS_k) + (1 - q)v_{k+1}(dS_k)], g(S_k)\}$

American Put

		0
	0.4	
1.36		1
	3	
		4

Hedging Portfolio : Δ

	$\Delta_1(up)$	
	-0.0833	
Δ_0		
-0.4333		
	-1	
	$\Delta_1(down)$	

Bank account : ψ

	0.85333	
3.09333		
	4	

Hedging Portfolio for American Option

Because, American options can be exercised early, the self-financing replicating portfolio is not exact.
Instead, the dynamics for replicating portfolio for American options is given as

$$C_k \geq 0$$

Handwritten notes:

$$\pi_k = (\Delta_k, \psi_k)$$

Arrows point from π_k to Δ_k and ψ_k . Δ_k points to S . ψ_k points to B . A red circle around k points to a red circle around $k+1$. A red circle around π_{k+1} is also present.

$$\pi_{k+1} = \Delta_k S_{k+1} + \left(\frac{B_{k+1}}{B_k} \right) (\pi_k - \Delta_k S_k - C_k)$$

$$\pi_{k+1} = \left(\frac{B_{k+1}}{B_k} \right) \pi_k + \Delta_k \left(S_{k+1} - \left(\frac{B_{k+1}}{B_k} \right) S_k \right) - \left(\frac{B_{k+1}}{B_k} \right) C_k$$

$$\mathbb{E} \left[\frac{\pi_{k+1}}{B_{k+1}} \right] = \mathbb{E} \left[\frac{\pi_k}{B_k} + \Delta_k \left(\frac{S_{k+1}}{B_{k+1}} - \frac{S_k}{B_k} \right) - \frac{C_k}{B_k} \right] \quad \text{standing at } k$$

Where $C_k \geq 0$ is the amount consumed.

Note that discounted portfolio process is a **super-martingale** under risk-neutral measure.

$$\mathbb{E}^{\mathbb{Q}} \left(\frac{\pi_{k+1}}{B_{k+1}} \middle| \mathcal{F}_k \right) = \left(\frac{\pi_k}{B_k} \right) + \Delta_k \left(\mathbb{E}^{\mathbb{Q}} \left(\frac{S_{k+1}}{B_{k+1}} \middle| \mathcal{F}_k \right) - \frac{S_k}{B_k} \right) - \frac{C_k}{B_k} = \frac{\pi_k}{B_k} - \frac{C_k}{B_k} \leq \frac{\pi_k}{B_k}$$

Handwritten notes: $\mathbb{E}[X_{k+1} | \mathcal{F}_k] \leq X_k$ and $(0, \gamma)$ with arrows pointing to the inequality.

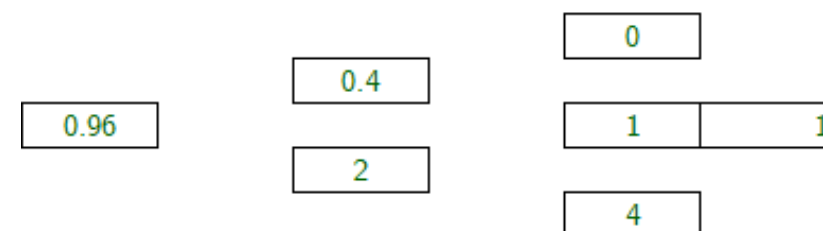
What is C_k ?

Whenever the American option is early-exercised, the value in that node will be higher than the continuation value. This is where the process **stops**. If we naively continue the risk-free growth from this point, it will result in an overestimated value of the derivative in the future nodes. Because of early exercise, the exact self-financing replication strategy breaks.

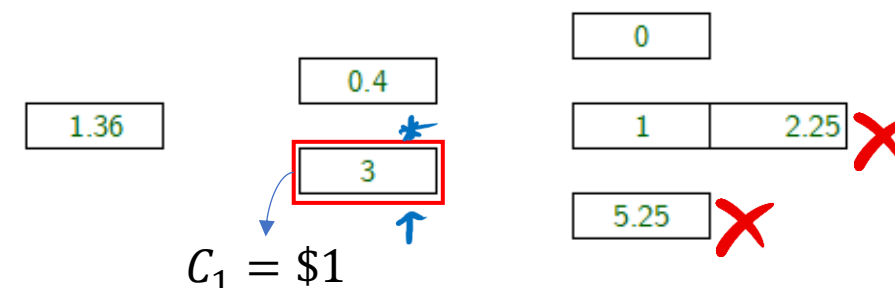


Whenever the **intrinsic value** > **continuation value**, if the option is not exercised (= continued), then the seller will consume the wealth C_k . Notice in our example, $C_1 = \$1$ for the node where the American option value exceeds the European counterpart by \$ 1.

(European Put)

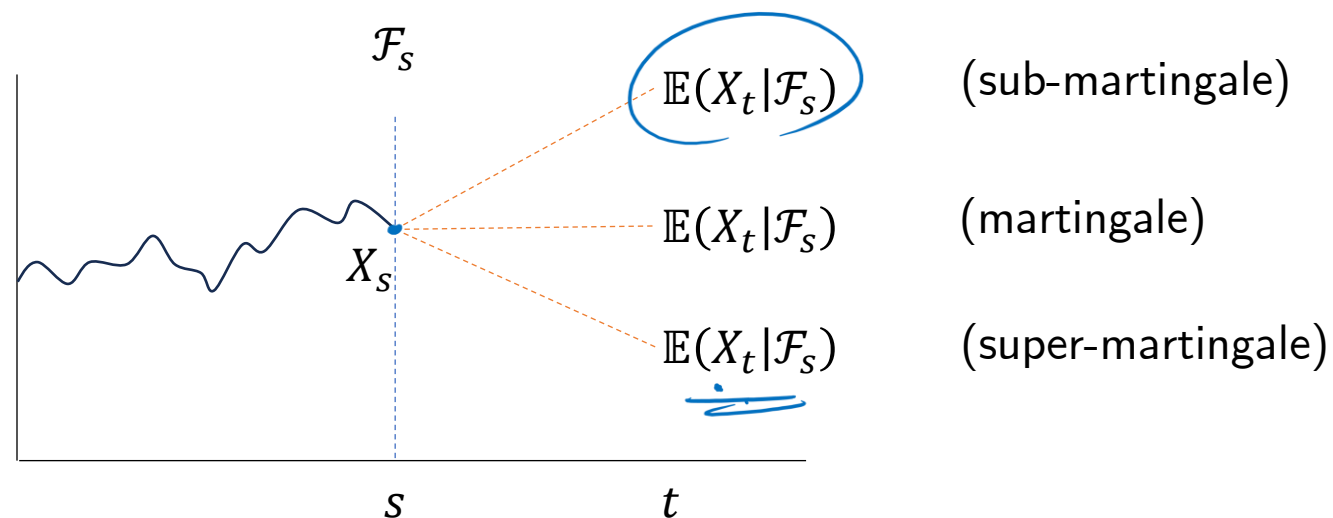


(American Put)



Martingale, Sub-Martingale, Super-Martingale

Below are the definitions of Martingales, Sub-Martingales and Super-Martingales



Martingale :

$$\mathbb{E}(X_t | \mathcal{F}_s) = X_s \quad (\text{no trend, drift}=0)$$

Sub-Martingale :

$$\mathbb{E}(X_t | \mathcal{F}_s) \geq X_s \quad (\text{trending up, drift} \geq 0)$$

Super-Martingale :

$$\mathbb{E}(X_t | \mathcal{F}_s) \leq X_s \quad (\text{trending down, drift} \leq 0)$$

Stopping-Time

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the probability space with filtration $\{F_k\}_{k=0}^n$. A **stopping time** is a random variable which is defined as $\tau: \Omega \rightarrow \{0, 1, 2, \dots, n\} \cup \infty$ with the property that

$$\{\omega \in \Omega; \tau(\omega) \leq k\} \in \mathcal{F}_k, \forall k = 0, 1, 2, \dots, n, \infty$$

In other words, τ is a stopping time if we know to stop at $\tau = t$ by using the information available till t . We do not need to look into the future. We can also say stopping time is $\mathcal{F}_t$ -adapted. Stopping time refers to a random time when we stop observing a process based on some rule.

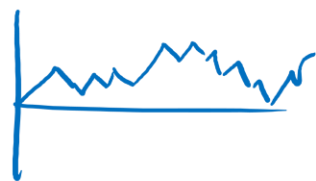
Applications of stopping times ✓



- ▶ American Option (Exercise time)
- ▶ Barrier Option (First passage times or hitting times)
- ▶ Credit Risk (firm value drops under a certain threshold)

Examples of Stopping time

Consider a simple symmetric random walk process $\{Z_t\}$ with $Z_0 = 0$



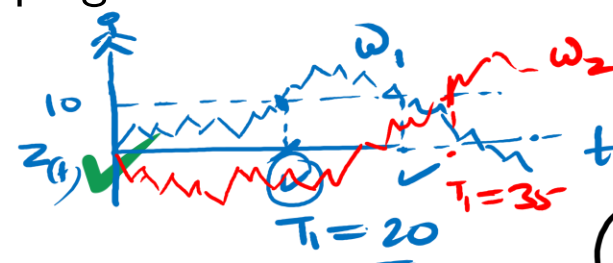
$$Z_t = \sum_{i=1}^n X_i$$

$$X_t = \begin{cases} 1, & \text{prob} = 0.5 \\ -1, & \text{prob} = 0.5 \end{cases}$$

$$\tau = k \in \mathcal{F}_k$$

Which of the following is a stopping time ?

I) $T_1 = \min \{t: Z_t = 10\}$



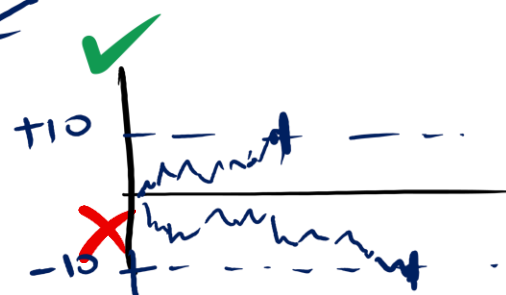
$$T_1(\omega_1) = 20$$

$$T_1(\omega_2) = 35$$

$$(Z_1, Z_2, \dots, Z_{20})$$

II) $T_2 = \min \{t: |Z_t| = 10\}$

III) $T_3 = \max \{t: |Z_t| = 10\}$



$$|Z| = 10$$

$$T_2 = 5$$

$$T_3 = 6$$

Important Results

Every constant time is trivially stopping time. In addition, if τ and θ are two stopping times, then

- 1) The minimum $t \wedge \tau := \min(\tau, \theta)$ is also a stopping time
- 2) The maximum $t \vee \tau := \max(\tau, \theta)$ is also a stopping time

$$x \wedge y = \min(x, y)$$

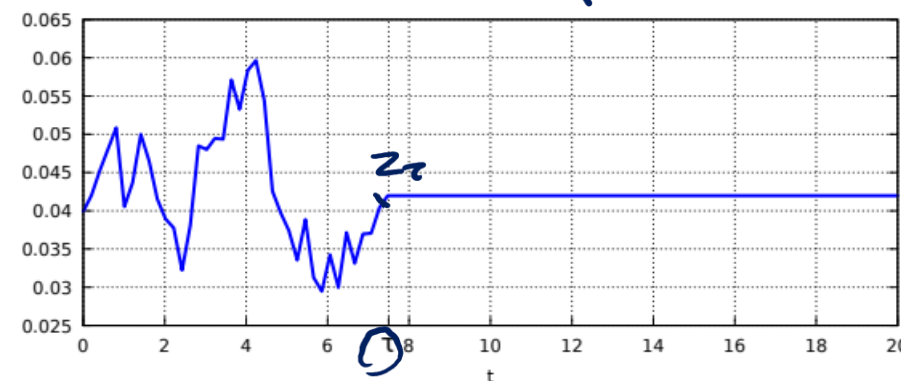
$$x \vee y = \max(x, y)$$

A stopped process $Z_{t \wedge \tau}$ is defined as

$$Z_t$$

$$Z_t^\tau = Z_{t \wedge \tau} = \begin{cases} Z_t, & t \leq \tau \\ Z_\tau, & t > \tau \end{cases}$$

Using indicator function we can also write $Z_{t \wedge \tau} = Z_t 1_{\{t \leq \tau\}} + Z_\tau 1_{\{t > \tau\}}$



Martingale with Stopping time

Consider a simple symmetric random walk process $\{Z_t\}$ with $Z_0 = 0$. We know that Z_t is a martingale for any deterministic time t , i.e.,



$$\mathbb{E}[Z_t] = \mathbb{E}[Z_0] = z_0$$

$$\mathbb{E}[Z_t | \mathcal{F}_s] = \mathbb{E}[z_s + X_{s+1} + X_{s+2} + \dots + X_t | \mathcal{F}_s]$$

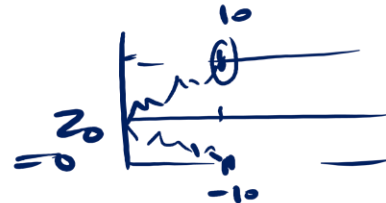
$$f_s = z_s +$$

But how about random stopping times?

$$\mathbb{E}[Z_\tau] = z_0 = ???$$

► I) $\tau = \min \{t: |Z_t| = 10\}$ → $\mathbb{E}[Z_\tau] = (10) \cdot \frac{1}{2} + \frac{1}{2}(-10) = 0 = z_0$

► II) $\rho = \min \{t: Z_t = 10\}$



$$\mathbb{E}[Z_\rho] = 10 \neq z_0$$

Doob's Optional Stopping Theorem (OST)

When random times are involved, we cannot automatically enforce the martingale property. There are certain regularity conditions which need to be met. It's called '**Doob's Optional Stopping**' theorem. The theorem states that

If τ is a random stopping time, then $\mathbb{E}[Z_\tau] = \mathbb{E}[Z_0]$ is true if any of the following conditions hold

- ▶ There exists a constant c such that for all t , $|Z_t| < c$ a.s Or
- ▶ There exists a constant c such $\tau < c$, a.s Or
- ▶ There exists a constant c such for all t , $\mathbb{E}[|Z_{t+1} - Z_t| \mathcal{F}_t] < c$ and $\mathbb{E}[\tau] < \infty$ a.s

To verify these conditions in our previous example, consider the stopped process Z_t^τ .

$$\mathbb{E}[Z_\tau^\tau] = \mathbb{E}[Z_\tau] = 0 = Z_0$$


$$Z_t^\tau = Z_{t \wedge \tau} = \begin{cases} Z_t, & t \leq \tau \\ Z_\tau, & t > \tau \end{cases}$$

$$t \leq \tau$$

$$|Z_t^\tau| < 1$$

$$\tau = \inf_{n \geq 1} \{Z_n = 1\}$$

Stopping time in American Option (I)

Let's define stopping time $\tau(\omega) = \min\{k; v_k(S_k) = (5 - S_k)^+\}$ which denotes the first time the option value agrees with the intrinsic value. Also, we will use the short-hand for the events

$$A_u = \{\text{first move is up}\} = \{uu, ud\}$$

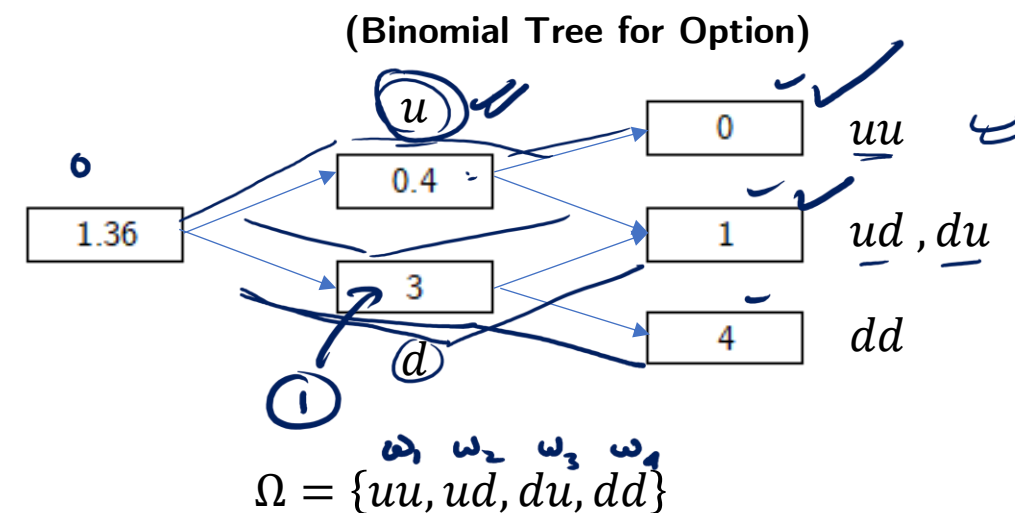
$$A_d = \{\text{first move is down}\} = \{du, dd\}$$

We note that
$$\tau(\omega) = \begin{cases} 1, & \text{if } \omega \in A_d \\ 2, & \text{if } \omega \in A_u \end{cases}$$

Also, we verify that τ is indeed a stopping time.

$(\tau \leq k) \in \mathcal{F}_k$

- $\rightarrow \{\omega; \tau(\omega) = 0\} = \phi \in \mathcal{F}_0$
- $\rightarrow \{\omega; \tau(\omega) = 1\} = A_d \in \mathcal{F}_1$
- $\rightarrow \{\omega; \tau(\omega) = 2\} = A_u \in \mathcal{F}_2$



$$\mathcal{F}_0 = \{\phi, \Omega\}$$

$$\mathcal{F}_1 = \{\phi, \Omega, A_u, A_d\}$$

$$\mathcal{F}_2 = \left\{ \phi, \Omega, A_u, A_d, uu, ud, du, dd, \text{ and all sets that can be created taking unions of these} \right\}$$

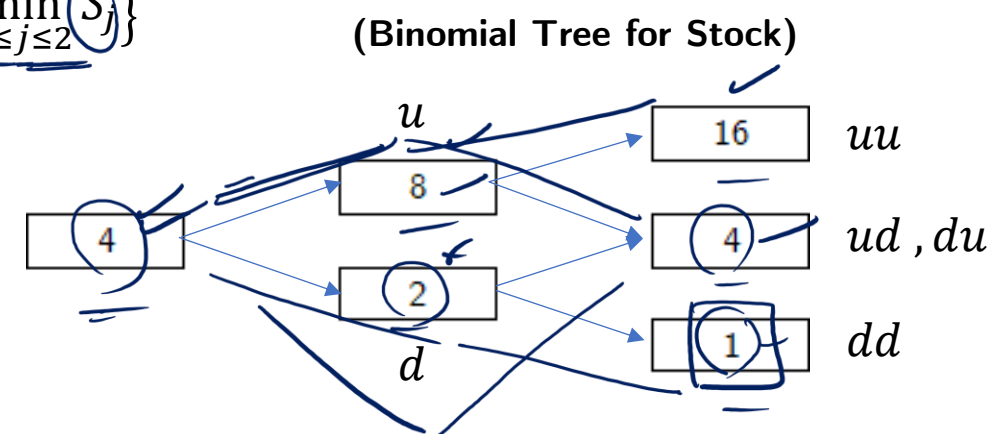
Stopping time in American Option (II)

Let's look at another random time on the tree that is NOT a stopping time. Define,

$$\rho(\omega) = \min_{0 \leq k \leq 2} \{k; S_k = \min_{0 \leq j \leq 2} S_j\}$$

We note that $\rho(\omega) = \begin{cases} 0, & \text{if } \omega \in A_u \\ 1, & \text{if } \omega = du \\ 2, & \text{if } \omega = dd \end{cases}$

$A_u = \{uu, ud\}$



Also, we verify that ρ is NOT a stopping time.

$$\begin{aligned} \{\omega; \rho(\omega) = 0\} &= A_u \notin \mathcal{F}_0 \quad \times \\ \{\omega; \rho(\omega) = 1\} &= \{du\} \notin \mathcal{F}_1 \quad \times \times \\ \{\omega; \rho(\omega) = 2\} &= \{dd\} \in \mathcal{F}_2 \quad \checkmark \end{aligned}$$

$$\Omega = \{uu, ud, du, dd\}$$

$$\mathcal{F}_0 = \{\phi, \Omega\} \quad \mathcal{F}_1 = \{\phi, \Omega, A_u, A_d\}$$

$$\mathcal{F}_2 = \left\{ \phi, \Omega, A_u, A_d, uu, ud, du, dd, \text{ and all sets that can be created taking unions of these} \right\}$$

Optional Sampling Theorem

The Optional Sampling theorem uses the same idea but for two random stopping times. So, if τ and ρ are two **bounded** stopping times, i.e., $\tau \leq c$, $\rho \leq c$, a.s where c is a constant, and

If $\tau \leq \rho$ a.s then,

$$\underline{\underline{\mathbb{E}[Z_\tau] = Z_0}}$$

$\boxed{Z_\tau = \mathbb{E}(Z_\rho \mathcal{F}_\tau)}$	(martingale)
$Z_\tau \leq \mathbb{E}(Z_\rho \mathcal{F}_\tau)$	(sub-martingale)
$Z_\tau \geq \mathbb{E}(Z_\rho \mathcal{F}_\tau)$	(super-martingale)

Optional Sampling Theorem on the Tree

To illustrate the idea of Optional Sampling theorem on the tree we define two topping times

$$\tau(\omega) = \min\{k; v_k(S_k) = (5 - S_k)^+\}$$

$$\rho(\omega) = 2$$

$$\tau = \rho$$

Under risk neutral probabilities the discounted stock price $Z_k = \left(\frac{4}{5}\right)^k S_k$ is a martingale.

Verify that for all paths ω

$$Z_\tau = \mathbb{E}(Z_\rho | \mathcal{F}_\tau)$$

$$Z_\rho = Z_2 = \left(\frac{4}{5}\right)^2 S_2$$

$$\left(\frac{4}{5}\right)^\tau S_\tau = \mathbb{E}\left(\left(\frac{4}{5}\right)^2 S_2 \mid \mathcal{F}_\tau\right)$$

(in other words)

$$\omega = uu, ud, du, dd$$